BRAC UNIVERSITY

Department of Computer Science and Engineering
CSE 341: Microprocessors

Question 1

Perform the following arithmetic operations and **store** the result in a variable called **"RESULT".**

RESULT = DA451Ch % F87h - 45 * 110010b + 62

Question 2

Two pet cats, **Kuro** and **Billu**, have their favorite sleeping spots marked by two numbers stored in registers. Due to a mischievous prank, their positions were accidentally interchanged.

Your task is to write an **8086 Assembly program** that swaps their positions back **without using any temporary register or memory variable**.

Initial Positions:

- **Kuro's Position** = Last non-zero digit of your Student ID
- **Billu's Position** = First non-zero digit of your Student ID
  (If both digits are the same, use the next non-zero digit for Billu's position.)

After Swapping:

- **Kuro's Position** = First non-zero digit of your Student ID *(or the next non-zero digit)*
- **Billu's Position** = Last non-zero digit of your Student ID